C++ Programming Exam - Answer Sheet

Student: Mike Wilson | ID: 2024003

Question 1 (15 points)

1.1 Which line of the above code is both a definition and an initialization?

Statement a

1.2 Which line of the above code declares a variable without allocating memory for it?

Statement c

1.3 Which lines of the above code are assignments?

Statement a and b

1.4 According to the following statement: const int *p;

both pointer p and the value of the integer that p points to cannot change

1.5 When is the following logical test true? (x>=y && ! x && x* y < 0 && y==0)

if x is greater than y, and y is equal to zero

1.6 Which of the following cases of mixed expressions is/are correct?

f / 3.33 is a float

1.7 Given: char c='b'; char *pc=&c; char *&rc;=pc; then after (*rc)++; what happens?

it stores 'b' in variable c

1.8 Which of the provided statements (if any) are invalid?

px = py;

1.9 Which of the provided statements would give the value of x?

*pp

1.10 Which expressions give as result an int equal to 6?

7 % 2, 29/5

1.11 What will be the value of x after: int x = (7>6 ? 1+8 : 8)

1

1.12 Which of the following(s) is a (are) valid function declaration?

void func(int){};

1.13 Which function will be called by: printFun(2.0*f) given float f;

void printFun(void)

1.14 How many times is function fib called when num is 3?

1

1.15 In a function with return type void, what happens at return?

0 is returned

Question 2 (5 points)

2.1 77 + c + i + 1L

char

2.2 6.55f + f / 1.5 - 9 / 8

float

2.3 'z' - 'z'

char

2.4 b + c

bool

2.5 't' - 'a' + c

char

2.6 77.8f + 4 * 0.5f + 45L

long

2.7 42L + (int) d + 94.3f + int(4.9)

int

2.8 0.0 + f + c

float

2.9 5.28L * d * 3 + 4.5

double

2.10 1.5f / d * f + 6.9 * 4L

float

Question 3 (10 points)

Replace the comments with expressions:

// Local name: cout << "\n Local name : " << local_name; // Global name: cout << "\n Global name : " << global_name; // Object member name: cout << "\n Object name : " << m->name; // Invoke member function print(): m->print();

Question 4 (10 points)

4.1 Fragment 1 output

x is 0 *y is 0 **z is 0

4.2 Fragment 2 output

a is 0 b is 0 c is 0 x is 0 y is 0 z is 0

4.3 Fragment 3 output

x[0] = 1 y[0] = 1

4.4 Fragment 4 output

test1=0test2=0test3=0

4.5 Fragment 5 output

At the beginning a = 10 b = 20 Before func2 a = 10 b = 20 After func2 a = 10 b = 20 At the end a = 10 b = 20

Question 5 (40 points)

Write a program that prints an hourglass figure:

#include using namespace std; int main() { int n; cin >> n; for (int i = 0; i < n; i++) { for (int j = 0; j < n; j++) { cout << '*'; } cout << endl; } return 0; }

Question 6 (20 points)

Provide the missing constructor and methods:

Complex() { real = 0; imaginary = 0; } int Complex::get_real() { return real; } int Complex::get_imaginary() { return imaginary; } void Complex::set_real(int r) { real = r; } void Complex::set_imaginary(int i) { imaginary = i; }